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import numpy as np
import cv2
import os
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_olivetti_faces
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

path = r"C:\Users\CAD LAB 17\Desktop\SE 07 ALML\olivetti_faces.npy"

if os.path.exists(path):
    imgs = np.load(path)
    labels = np.array([i // 10 for i in range(len(imgs))])
else:
    data = fetch_olivetti_faces()
    imgs, labels = data.images, data.target

print("Dataset:", imgs.shape, "| Classes:", len(np.unique(labels)))

imgs = np.array([
    cv2.cvtColor((i * 255).astype("uint8"), cv2.COLOR_GRAY2BGR)
    for i in imgs
])

features = np.array([
    np.hstack([
        cv2.normalize(
            cv2.calcHist([im], [c], None, [256], [0, 256]),
            None
        ).flatten()
        for c in range(3)
    ])
    for im in imgs
])

print("Feature Shape:", features.shape)

Xtr, Xte, Ytr, Yte, Imtr, Imte = train_test_split(
    features, labels, imgs, test_size=0.2, random_state=42,
    stratify=labels
)

model = RandomForestClassifier(n_estimators=200, random_state=42)
model.fit(Xtr, Ytr)

pred = model.predict(Xte)

print("Accuracy:", round(accuracy_score(Yte, pred) * 100, 2), "%")

plt.figure(figsize=(10, 8))

for i in range(6):
    plt.subplot(2, 3, i + 1)

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plt.imshow(cv2.cvtColor(Imte[i], cv2.COLOR_BGR2RGB))
plt.title(f"A:{Yte[i]} | P:{pred[i]}")
plt.axis("off")

plt.suptitle("Color Face Recognition Output")
plt.tight_layout()
plt.show()
```